

Pairings

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Embedding degree

- Given elliptic curve $\mathbb{E}$ over $\mathbb{F}_q$, a large prime 'r' dividing the order of $\mathbb{E}(\mathbb{F}_q)$, the embedding degree k is the smallest positive integer —

$$r | q^k - 1$$

- In other words, k is the smallest integer for which $\mathbb{F}_{q^k}$ contains the r^{th} root of unity.
- 'k' determines
 - Target field G_T where pairings land
 - Security level against index calculus attacks (on G_T)

- In pairings over elliptic curve:

$$e : G_1 \times G_2 \rightarrow G_T$$

- G_1, G_2 are subgroups of $\mathbb{E}(\mathbb{F}_q)$
- $G_T \subseteq \mathbb{F}_{q^k}^*$ (multiplicative group of $\mathbb{F}_q$'s extension field.)

Pairings and field relation

Type	G_1	G_2	G_T	k
Symmetric (Weil / Tate)	$\mathbb{F}_q$	$\mathbb{F}_q$	$\mathbb{F}_{q^k}^*$	smallest k : $r q^k - 1$
Asymmetric (BLS)	$\mathbb{F}_q$	$\mathbb{F}_{q^d}$ (twist of $\mathbb{E}$)	$\mathbb{F}_{q^k}^*$	smallest k: $r q^k - 1$ and $d k$

Notes

- G_1 and G_2 are groups from the elliptic curves, not the whole EC.
- We cannot use G_T group values as x on an EC.
- For pairings, the two input points must be from groups of the same prime order 'r'. If $P \in G_1, Q \in G_2$, then $rP = O_1, rQ = O_2$.
- If not, then we lose bilinearity:

$$e(aP, Q) \neq e(P, aQ)$$
- We also lose non degeneracy, $e(P, Q) = 1$

Examples

Curve	k (embedding degree)	examples
BLS curve: (Barreto Lynn Scott)	12, efficient pairing	BLS12-381, 48 bytes, 3 bits used for flags
BN curve (Barreto Naehrig)	12	zk-snarks, bn254
KSS (Kachisa, Schaefer, Scott)	high k = 18, 24	used when longer security parameters are needed

Pairing over symmetric groups (to do)

$G_1 \rightarrow$, additive group of order r with additive identity $\mathcal{O}$.

$G_T \rightarrow$, multiplicative group of order r with multiplicative identity 1 .

$e(G_1, G_1) \rightarrow, G_T$

